

Nisip Sigdel

Computer Science Graduate | AI, Machine Learning & NLP

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SUMMARY

Computer Science graduate with hands-on experience in Natural Language Processing, Machine Learning, Large Language Models, and Retrieval-Augmented Generation. Developed AI systems involving transformer-based NLP models, semantic search, vector databases, and locally deployed LLMs. Interested in research focused on NLP, LLMs, information retrieval, and intelligent systems for understanding and processing real-world language and document data.

EDUCATION

B.Sc. Computer Science & Information Technology

Tribhuvan University

04/2022 – 03/2026

Kathmandu, Nepal

Completed all 8 semesters of the BSc CSIT program in 2026. Contributed to an undergraduate group project focused on NLP-based resume screening and job recommendation, with primary involvement in dataset preparation and transformer-based model training using spaCy, RoBERTa, and Google Colab.

RESEARCH INTERESTS

AI, Machine Learning & NLP

LLMs | RAG | Information Retrieval | Intelligent Systems

Research interests in Natural Language Processing, Large Language Models, Retrieval-Augmented Generation, Information Retrieval, and intelligent systems, with a focus on developing AI systems for understanding, retrieving, and processing real-world language and document data.

PROJECTS

AI PDF Reader & Autonomous Finance RAG Assistant

07/2026 – 08/2026

RAG · LLMs · NLP · Information Retrieval · Python · Ollama · ChromaDB

- Developed an integrated **Retrieval-Augmented Generation (RAG)** platform combining a local AI-powered PDF document assistant with an autonomous finance analysis system.
- Built a document retrieval pipeline using **PDF processing, text chunking, Hugging Face embeddings, ChromaDB/FAISS, and semantic search** for context-aware question answering.
- Integrated **Ollama with Llama 3.1 8B** for local LLM inference, enabling AI processing without external AI APIs.
- Developed an autonomous financial news pipeline using **Google News RSS, BeautifulSoup, yfinance, and scheduled background tasks** to collect and index current financial information.
- Designed retrieval and prompt constraints to keep responses grounded in retrieved context and prevent direct investment recommendations.
- Built a modular **Streamlit** interface with a PDF Assistant, Live Finance Dashboard, and standalone chat assistant, with **SQLite** for application logging and **Pytest** for testing.

NLP-Based Job Recommendation & Resume Screening System

NLP Model Training | spaCy Transformers | RoBERTa | Google Colab

- Contributed to an undergraduate group project focused on NLP-based resume screening and job recommendation.
- Prepared and processed training data and configured the NLP pipeline for model training.
- Trained a transformer-based Named Entity Recognition (NER) model using spaCy Transformers and RoBERTa-base in Google Colab.
- Tested and evaluated the trained model and successfully ran the saved model for resume entity extraction.

- Saved the trained model for potential integration into the broader application.

GameVault

Full-Stack Web Application | React | Node.js | Express | PostgreSQL

- Developed a full-stack web application for sharing and discovering links to **legally distributable community-created game resources**, including patches, save files, custom maps, and mods.
- Built the frontend using **React and Vite**, with a modular component architecture and **mobile-first responsive design**.
- Developed backend services using **Node.js and Express**, with **PostgreSQL hosted on Railway** for database management and API integration.
- Implemented **user authentication, global search and filtering, comments, ratings, and an administrative dashboard** to support community interaction and content management.
- Integrated **Cloudinary** for dynamic image handling and deployed the frontend on **Vercel**.
- Implemented **interactive UI elements, smooth transitions, and a custom particle-effect cursor** to enhance the overall user experience.

PROFESSIONAL EXPERIENCE

Co-Founder | Web Developer | AI SEO & AEO

DigitalIdea2Grow 

07/2022 – Present
Biratnagar, Nepal

- Co-founded DigitalIdea2Grow, an AI SEO and digital marketing company focused on web development, AI-driven content, and search optimization.
- Developed responsive websites and contributed to client requirements, project planning, and user experience improvements.
- Researched and developed content related to **Artificial Intelligence and emerging AI technologies**, exploring their applications in search, content, and digital marketing.
- Researched and optimized AI SEO and Answer Engine Optimization (AEO) content to improve visibility across traditional search engines and AI-powered answer engines.

Software Developer (Freelance)

Mansa Tech Group 

04/2024 – Present
Remote

- Developed responsive web applications and user-focused interfaces for business clients.
- Built and integrated frontend components with backend services through **REST APIs** for dynamic data exchange and application functionality.
- Worked across frontend and backend development to deliver complete web-based solutions tailored to client requirements.
- Collaborated on application development and deployment while troubleshooting issues and improving system functionality and reliability.

Quality Assurance Intern

Elite Technologies 

10/2025 – 12/2025
Biratnagar, Nepal · Remote

- Analyzed application requirements and designed test cases covering **UI, navigation, form validation, and core functionality**.
- Executed **manual, regression, and cross-browser testing** to identify functional and usability issues.
- Documented and reported software defects using **Jira**, including reproduction steps, expected and actual results, and severity.
- Maintained test case documentation and testing records using **Microsoft Excel**.
- Collaborated with development and design teams to communicate issues and support software quality improvements.

SKILLS

AI, Machine Learning & NLP

Natural Language Processing (NLP) · Machine Learning · Large Language Models (LLMs) · Retrieval-Augmented Generation (RAG) · Information Retrieval · Semantic Search · Transformer Models · spaCy · RoBERTa

Full-Stack Web Development

React.js · Node.js · Express.js · PostgreSQL · REST APIs · HTML · CSS · JavaScript

Other

SEO · AEO · WordPress · C++ · C# · C

AI & Data Tools

Python · LangChain · Ollama · Llama 3.1 · ChromaDB · FAISS · Hugging Face Embeddings · Streamlit · SQLite · Pytest

Software Testing & Quality Assurance

Manual Testing · Functional Testing · Regression Testing · Test Design · Test Execution · Cross-Browser Testing · Bug Tracking · Jira

CERTIFICATES

Python Data Structures

University of Michigan · Coursera · March 2023

Programming for Everybody

University of Michigan · Coursera · January 2023

LANGUAGES

- English
 - Hindi
- Nepali